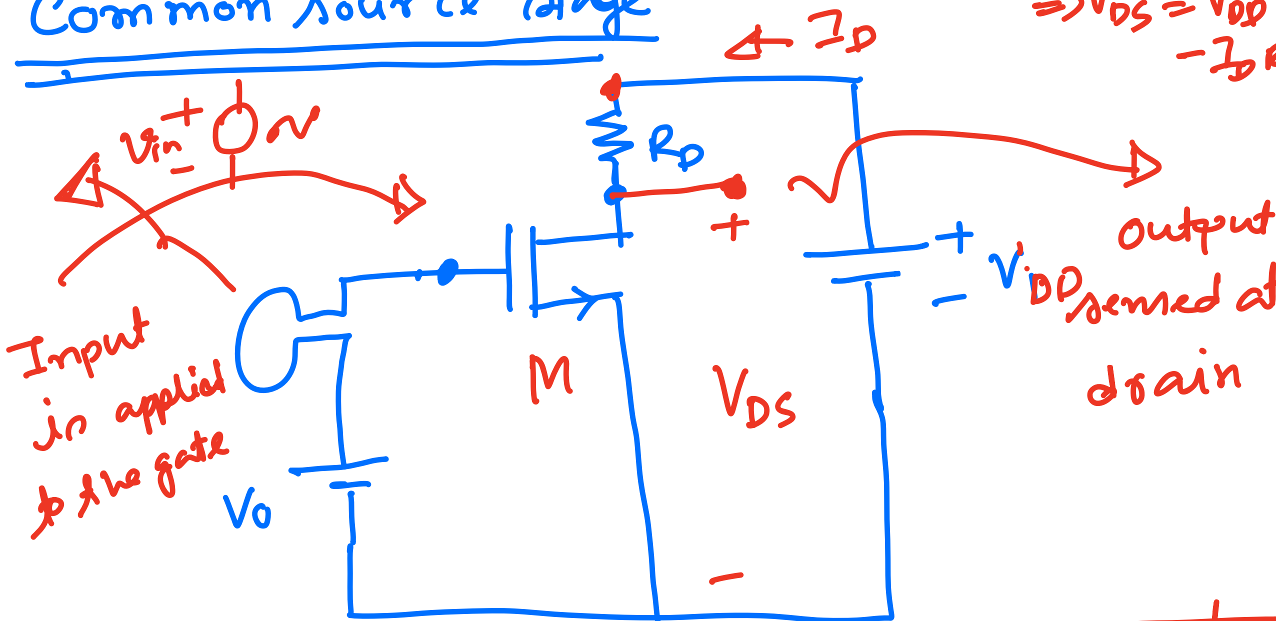
CMOS Amplifier

Common source stage

$V_{DD} = I_D R_D + V_{DS}$

$\Rightarrow V_{DS} = V_{DD} - I_D R_D$



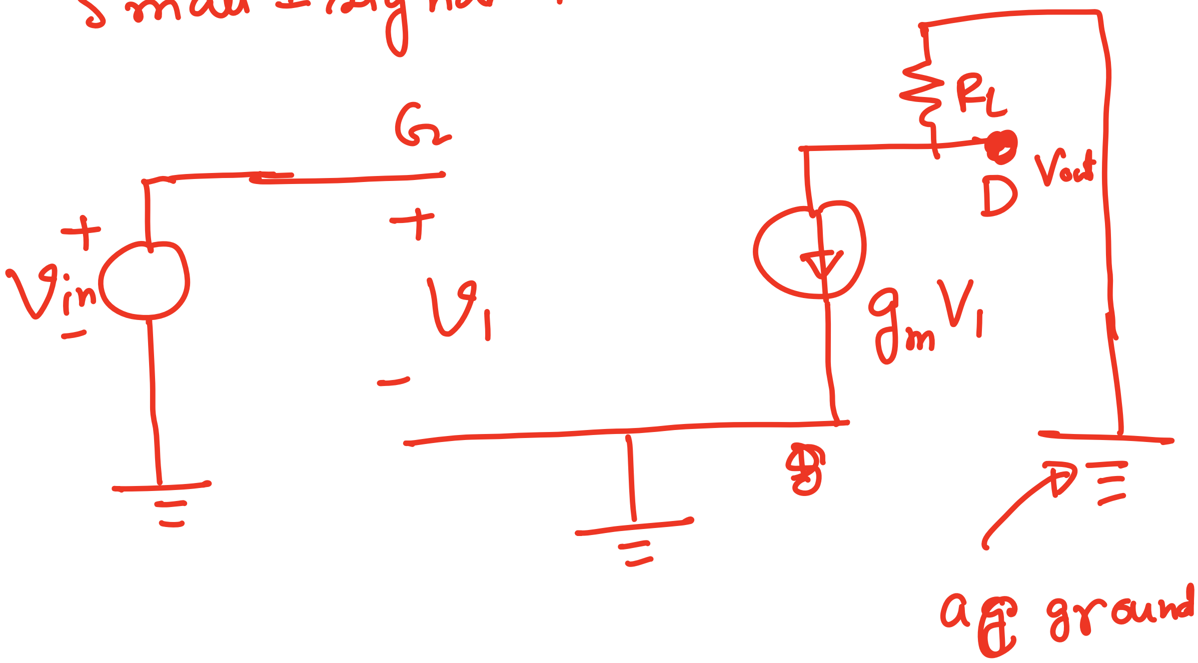
For M_1 to maintain sat:

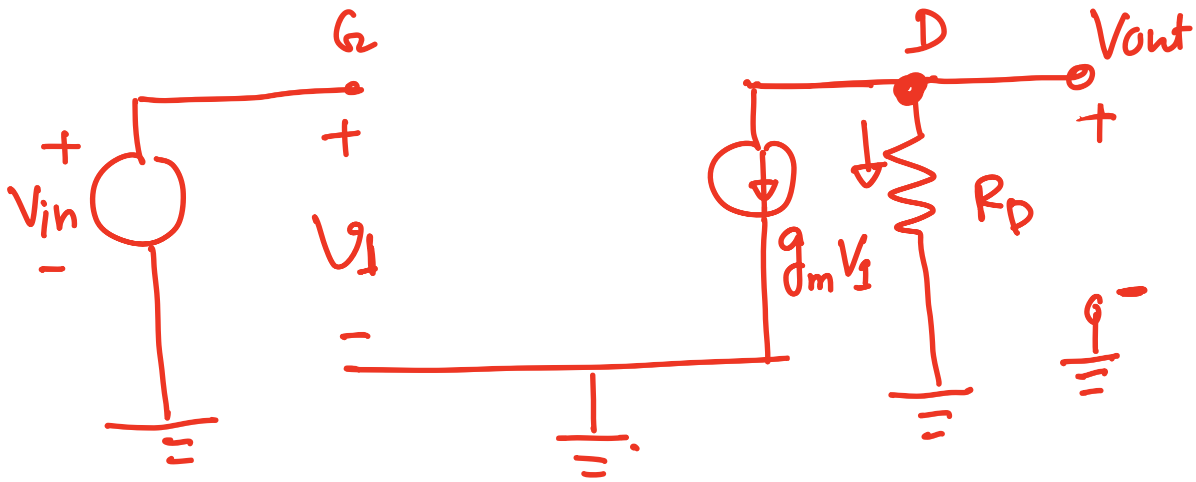
$V_{DS} > V_{GS} - V_{Th}$

$\Rightarrow V_{DD} - I_D R_D > V_{GS} - V_{th}$

A_V

Small-signal Model.





$$V_1 = V_{in}$$

$$g_m V_1 = g_m V_{in}$$

KCL @ node D,

$$g_m V_{in} + \frac{V_{out}}{R_D} = 0$$

$$\therefore \frac{V_{out}}{V_{in}} = -g_m R_D$$

$$\therefore \boxed{A_V = -g_m R_D}$$

when $\lambda = 0$

$A_V = -g_m \times$ (total resistance tied between drain and ac GND)

*Q: Draw the circuit diagram of the CS stage and find the small signal gain (Consider $\lambda = 0$)

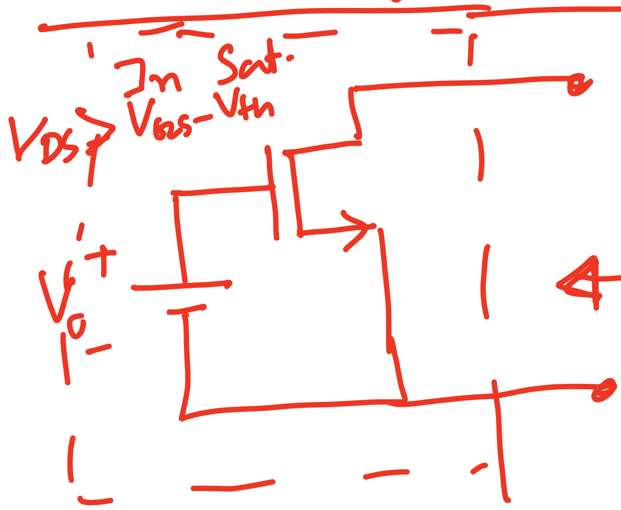
Example 7.4

$\lambda \neq 0$

$$A_v = -g_m (R_D \parallel r_o)$$

CS stage with current-Source Load

Realization of current Source:



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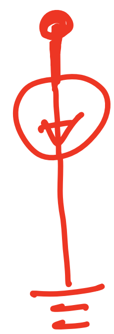
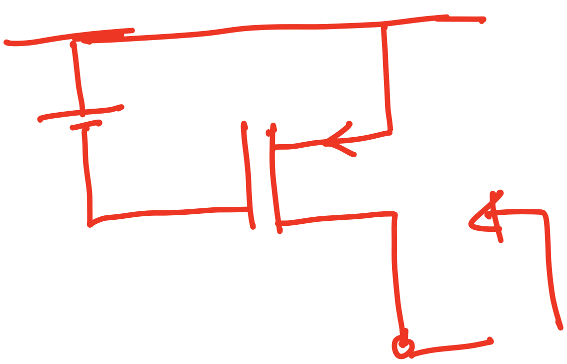
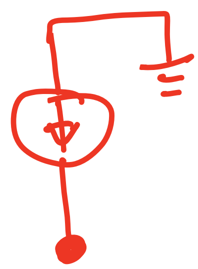


Small signal Model?

out of the node

N-MOS current source



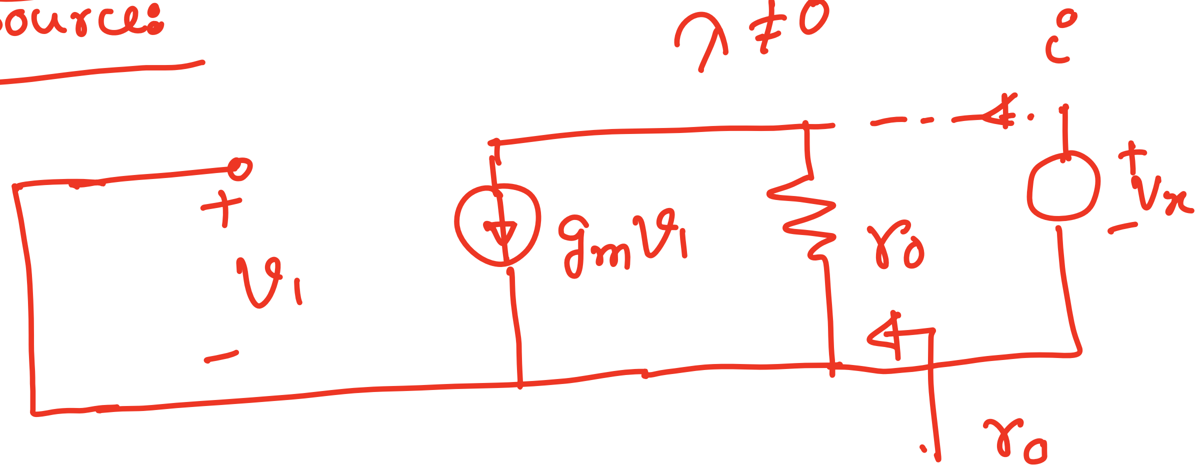


into the node

P-MOS

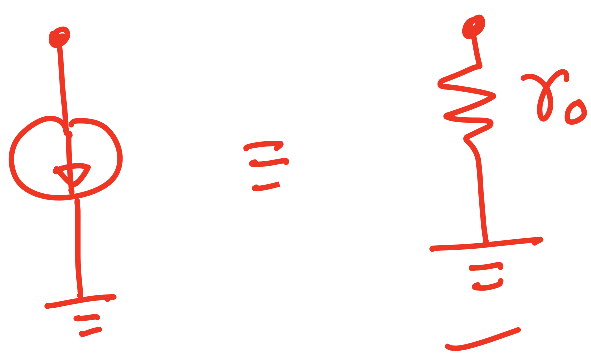
Small signal Model of MOS current Source:

$\lambda \neq 0$

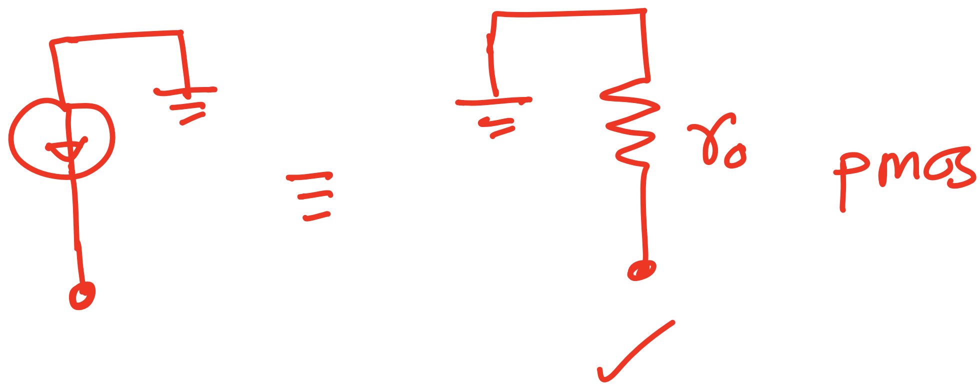


$$\frac{v_x}{i_x} = r_0$$

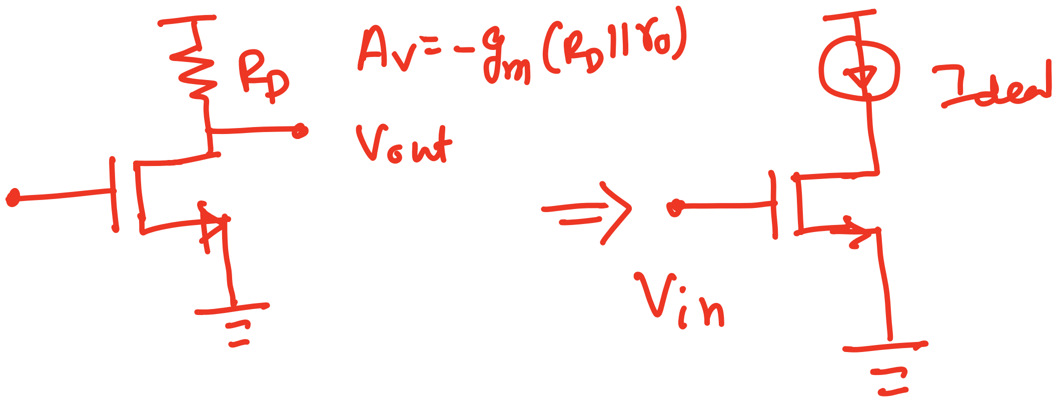
So small signal model of MOS current source (if $\lambda \neq 0$)



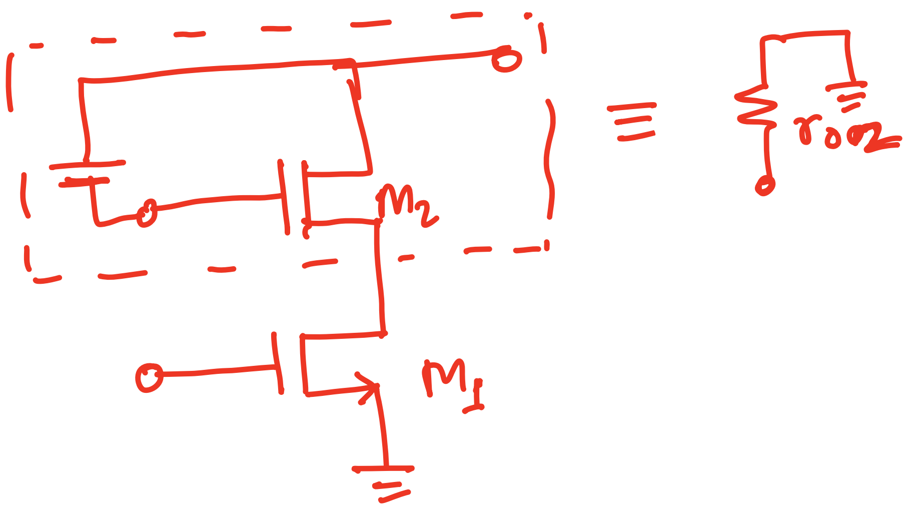
NMOS

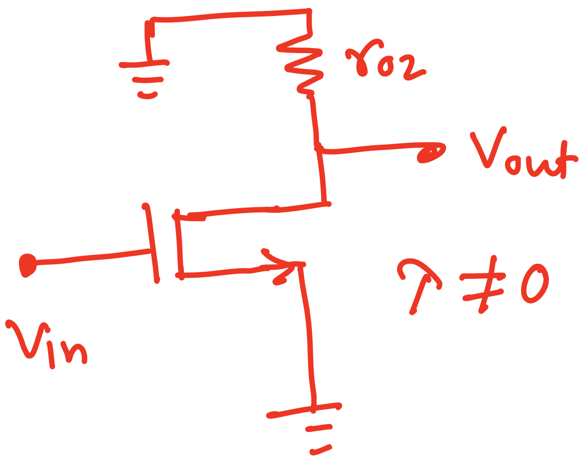


$A_v = -g_m (R_D \| r_o)$



Example:





$\lambda \neq 0$

$$\boxed{A_v = -g_{m1}\, r_{o2} \| r_{o1}}$$

Example 7.6